



CHAPTER

Valves and Manifold

Automated 3-way valves and the feed distribution manifold

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11.1 Overview and function

Automated 3-way valves direct flow between feed, product and recirculation paths (e.g. for electrodialysis reversal, chapter 10), while the manifold assembly distributes feed water evenly to every stack on a skid.



Figure 11.1. Suggested figure: photograph of the manifold assembly with its 3-way valves, labeled by function (feed, product, recirculation).

11.2 Key specifications

- 3-way automated valves: PVC construction.
- Manifold: DN25–DN63, PVC and stainless-steel fittings.

Two actuated-valve product lines are confirmed present in HydroVolta's component library as candidates for this duty:

- **Georg Fischer (+GF+) e-DIASTAR Type 514** actuated ball valve: PVC-U body, 24 V AC/DC multiturn actuator with integrated open/closed position feedback, built-in heating element and manual emergency override. Available in DN25 (1", PN10) and DN50 (2", PN6) sizes matching this manual's connection sizes elsewhere; seal options EPDM or PTFE/EPDM.
- **Bürkert Type 3003** electric rotary actuator, fitted to quarter-turn ball/butterfly valves: available as on/off or proportional control (accepting/outputting 4–20 mA, 0–20 mA or 0–10 V); standard manual override, adjustable limit switches, integrated heating resistor and torque limiter; UL 94 V0 flame-resistant housing.

Both product lines run on the same 24 V DC control bus documented for the cabinet (chapter 14), consistent with the troubleshooting guidance below.

11.3 Operation

Valves are switched automatically by the PLC as part of the stream- reversal and CIP sequences (chapters 6 and 10). TODO: no document describes manual valve override procedure or fail-safe position on power loss — both candidate actuator families above include a manual override lever, but the fail-safe (last-position vs. defined) behavior on power loss has not been confirmed.

11.4 Maintenance

See chapter 7 for the general schedule and log template.

11.5 Troubleshooting

- **Valve not switching:** check 24 V DC control voltage at the valve terminal against PE before assuming a valve fault (chapter 14).
- **Flow imbalance across stacks:** check manifold for blockage or a partially closed valve before suspecting a pump or stack issue.